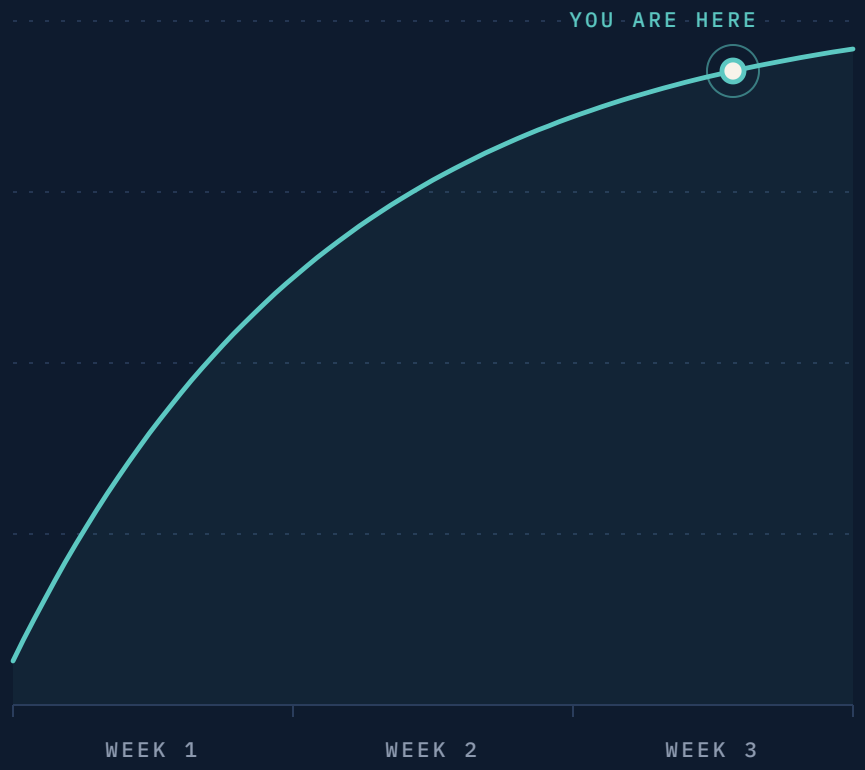
WEEK 3 · DAY 3 · OVERVIEW

Hierarchical Clustering and Cluster Profiling



Created by **Jeff Lenamon**

ABOUT 50 MIN WITH EXERCISES / 12 SECTIONS / 4 EXERCISES

This course is educational content created in a personal capacity. Nothing here is investment advice or a recommendation to buy or sell any security. All portfolio data is synthetic.

Problem Statement

BUSINESS CONTEXT

The 821 bonds of the benchmark come from 150 issuers.

A credit team wants the issuers grouped by spread, duration, size and rating.

Before choosing how many groups, it wants to see the whole family tree of who resembles whom.

Then it wants one readable line per group, in the desk's own units.

The bonds and issuers are invented for this course.

OBJECTIVES

Build one row per issuer, join the issuers from the bottom up into a tree, read the tree and cut it, compare four ways of joining, and describe each cluster in bp, years and billions beside Day 12's K-means.

ABOUT 50 MIN WITH EXERCISES / 12 SECTIONS / 4 EXERCISES

Your Route Through Day 3

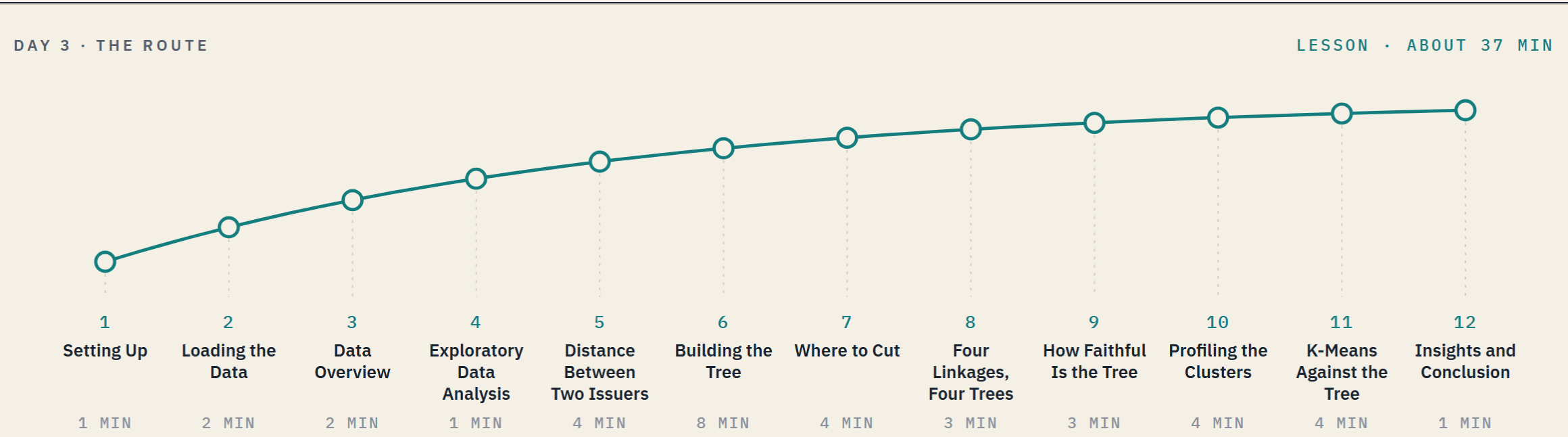


FIGURE 1 Twelve sections, in order, with about how long each takes.

Time. About 37 minutes for the lesson, then about 12 for the four exercises.

Fix the seed and load the libraries.

Setting Up

FIX THE SEED, FIND THE DATA

```
import os
from pathlib import Path
os.environ["OMP_NUM_THREADS"] = "1"
os.environ["OPENBLAS_NUM_THREADS"] = "1"
SEED = 42
if not Path(DATA + "benchmark.csv").exists():
```

WHAT THE CELL DOES

Run this cell and move on: it fixes the seed and finds the course data.

LOAD THE LIBRARIES

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from ipywidgets import interact
from scipy.spatial.distance import pdist, squareform
from sklearn.preprocessing import StandardScaler
from sklearn.cluster import AgglomerativeClustering, KMeans
from sklearn.metrics import silhouette_score
sns.set_theme(style="whitegrid")
sns.set_palette("colorblind")
plt.rcParams["figure.figsize"] = (10, 5)
```

Loading the Data

Bezane Finance (BEZE): six bonds, BBB-

OAS (bp)	duration (years)	amount (billions)
147	4.627	1.65
172	7.824	0.95
178	6.260	0.85
188	1.382	0.95
189	7.238	2.45
215	9.462	2.45

sorted by OAS: the two shaded rows hold the middle spreads

groupby →

one row for BEZE

median_oas	**183.0 bp** median of the six
median_duration	**6.749 years** median of the six
bonds	**6** how many
amount_bn	**9.30 billion** sum of the six

the median: one odd bond barely moves its issuer's row

FIGURE 2 Bezane Finance's six bonds become one row: 183.0 bp, 6.749 years, a count of six and 9.30 billion.

READ THE THREE FILES

```
bonds = pd.read_csv(DATA + "benchmark.csv")
ratings = pd.read_csv(DATA + "ratings.csv")
names = pd.read_csv(DATA + "issuers.csv")
bonds["log_oas"] = np.log(bonds["oas"])
```

WHY MEDIANS

The medians keep one unusual bond from moving its issuer's row much.

Each issuer's bonds collapse into one row: the median spread and duration, how many bonds, and their total size.

Check the rows, ranges and missing values.

Data Overview

FIVE COLUMNS ON FIVE SCALES

	MEAN	STD	MIN	50%	MAX
median_oas	171.52	144.23	21.00	120.50	922.50
median_log_oas	4.88	0.71	3.04	4.79	6.83
median_duration	5.85	1.93	1.02	5.85	13.07
bonds	5.47	1.74	3.00	5.50	8.00
amount_bn	7.62	2.79	1.70	7.48	15.00
score	9.21	3.53	1.00	9.00	18.00

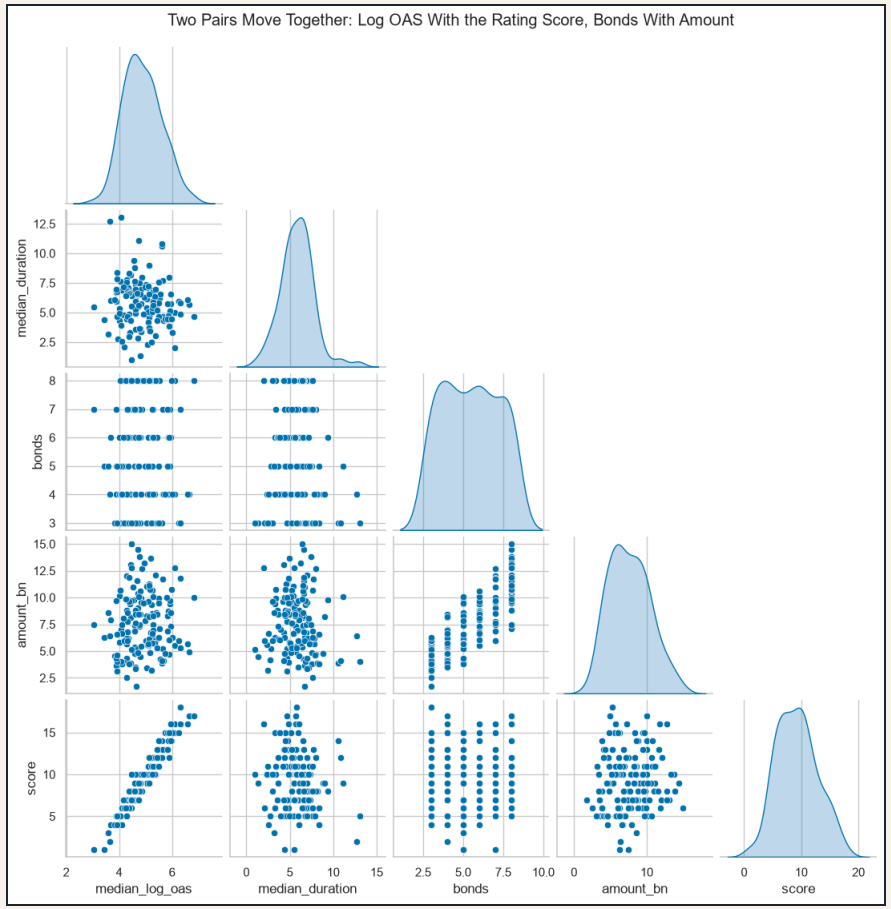
WHAT THE RANGES SHOW

Median OAS runs from 21 to 922.5 bp, and every issuer has 3 to 8 bonds.

The total size runs from 1.70 to 15.00 billion, and the score covers the whole scale, 1 to 18.

150 clean issuers, five columns on five scales. Day 12 showed what that does to a distance, so we will scale.

Exploratory Data Analysis



DRAW EVERY PAIR ONCE

```
sns.pairplot(issuers[features], corner=True, diag_kind="kde", height=2.2)
```

THE TWO TIGHTEST PAIRS AS CORRELATIONS

```
0.97 0.82
```

READING THE PAIRS

Log OAS and the rating score lie almost on a line: correlation 0.97.

Two columns say credit and two say size, so a distance will hear credit twice. We keep that in mind and measure distance.

Distance Between Two Issuers

ONE STANDARD DEVIATION OF EACH COLUMN

```
median_log_oas     0.71
median_duration    1.93
bonds              1.73
amount_bn          2.78
score              3.52
dtype: float64
```

THE GAP BETWEEN THE FIRST TWO ISSUERS

```
Distance: 2.0227
```

EVERY PAIR AT ONCE

```
pair_distances = pdist(X_scaled)
```

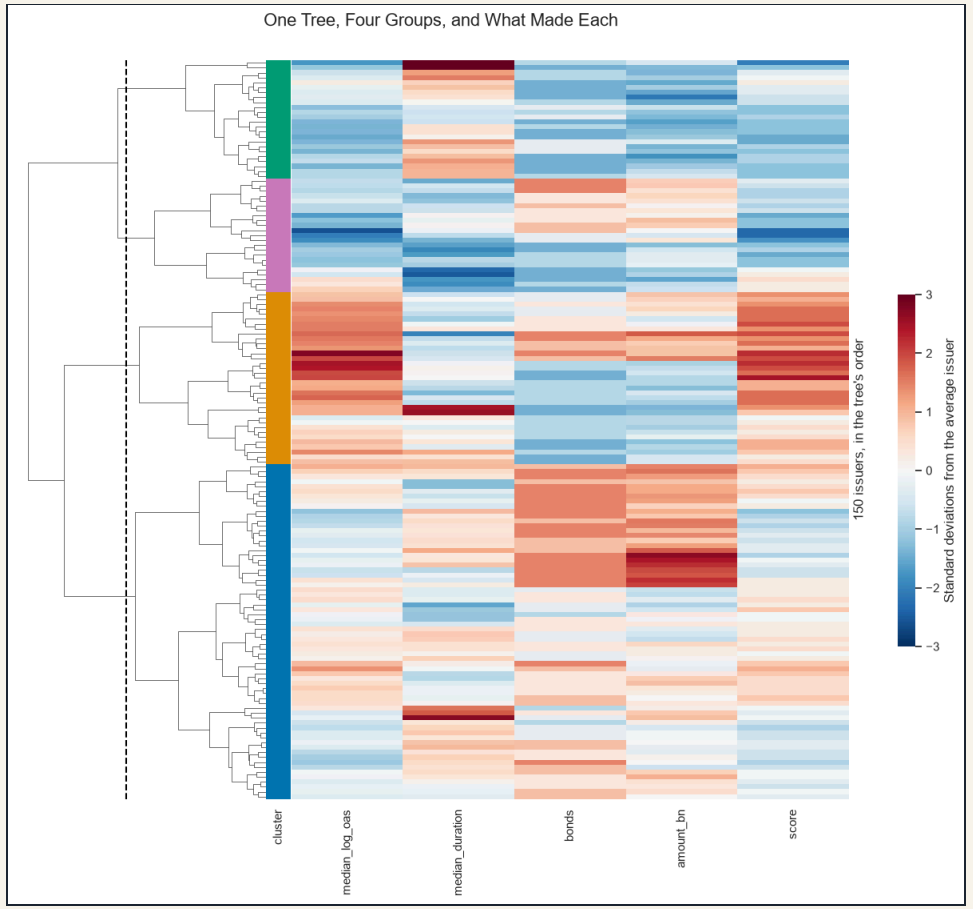
```
11175 pairs; median 2.86 ; largest 8.31
```

THE DISTANCE TABLE

TICKER	BEZE	BEZH	BEZL	BEZN
ticker				
BEZE	0.0000	2.0227	1.2449	1.3783
BEZH	2.0227	0.0000	1.8432	2.5416
BEZL	1.2449	1.8432	0.0000	1.7445
BEZN	1.3783	2.5416	1.7445	0.0000

BEZE and BEZH are 2.0227 apart, most of it from log OAS and the score: credit, heard twice.

Building the Tree



CUT THE TREE AT FOUR CLUSTERS

```
by_count = fcluster(merges, t=4, criterion="maxclust")
by_height = fcluster(merges, t=cut, criterion="distance")
print("Same labels:", (by_count == by_height).all())
```

```
Same labels: True
```

READING THE FOUR STRIPS

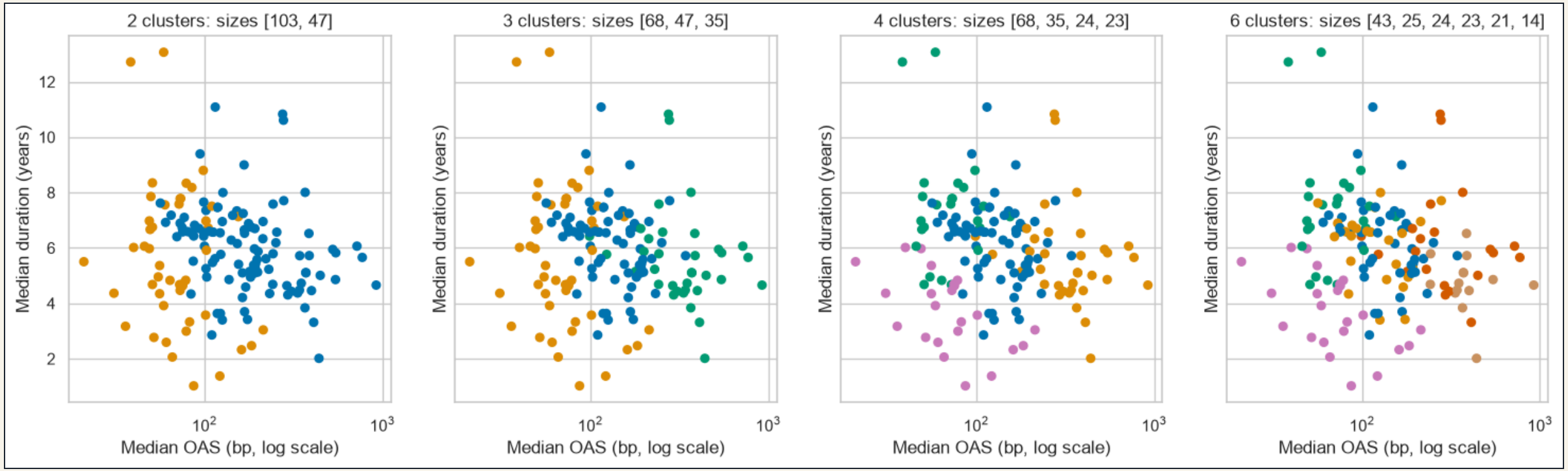
Cluster 1, the orange strip, is red on log OAS and the score: +1.28 on both.

Cluster 0, the blue strip, is red on bonds and amount, +0.64 and +0.58, and near 0 on credit.

Clusters 2 and 3, green and pink, are blue on credit, about -0.9 each, and split by duration: +0.81 against -1.12.

One tree, four groups, and you can see what made each. Next: why four, and not three or six.

Where to Cut



The widest band says 3 and the silhouette says almost nothing. We take 4, because the fourth cluster splits the tighter issuers by duration: a judgement, stated as one.

Four Linkages, Four Trees

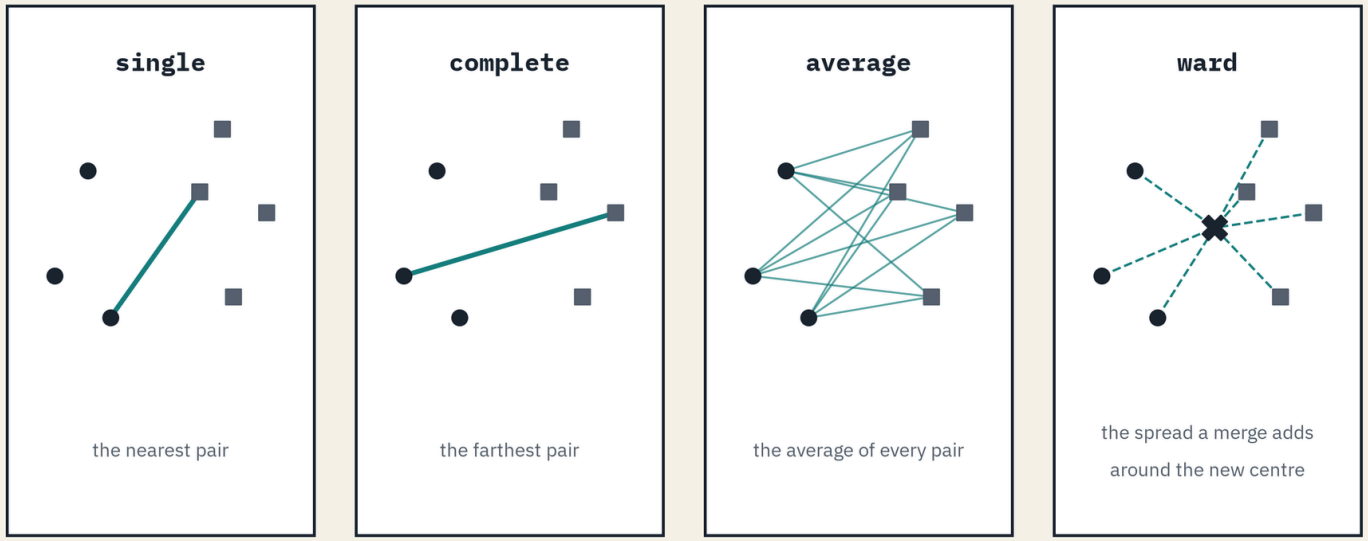


FIGURE 4 Single takes the nearest pair, complete the farthest, average every pair, ward the spread added around the centre.

FOUR CLUSTERS FROM TWO OF THE LINKAGES

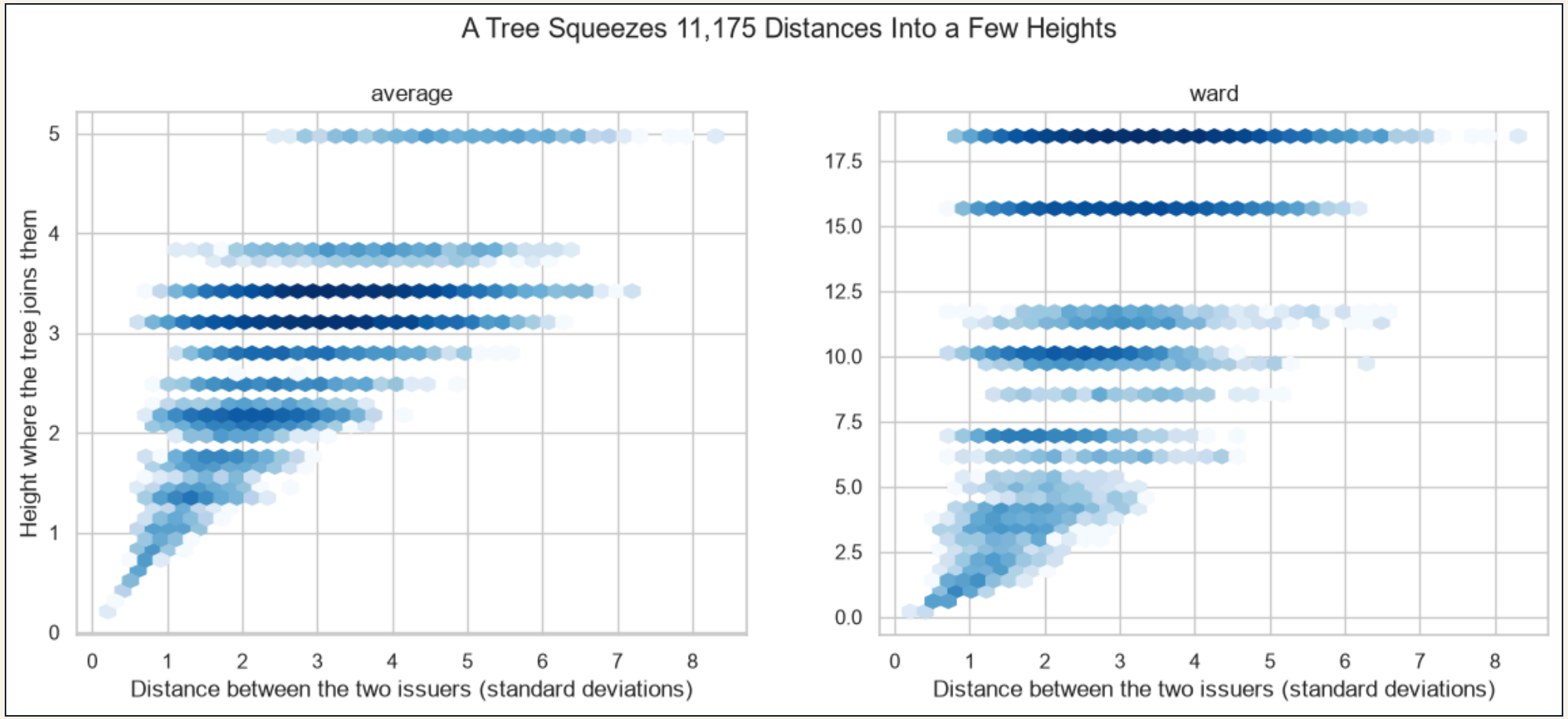
	COMPLETE	WARD
cluster 0	64	68
cluster 1	35	35
cluster 2	28	24
cluster 3	23	23

WHY A LINKAGE MATTERS

Ward is one rule for how close two clusters are; a **linkage** is that rule.

Same distances, four meanings of close, four answers. Next: which tree keeps the distances best.

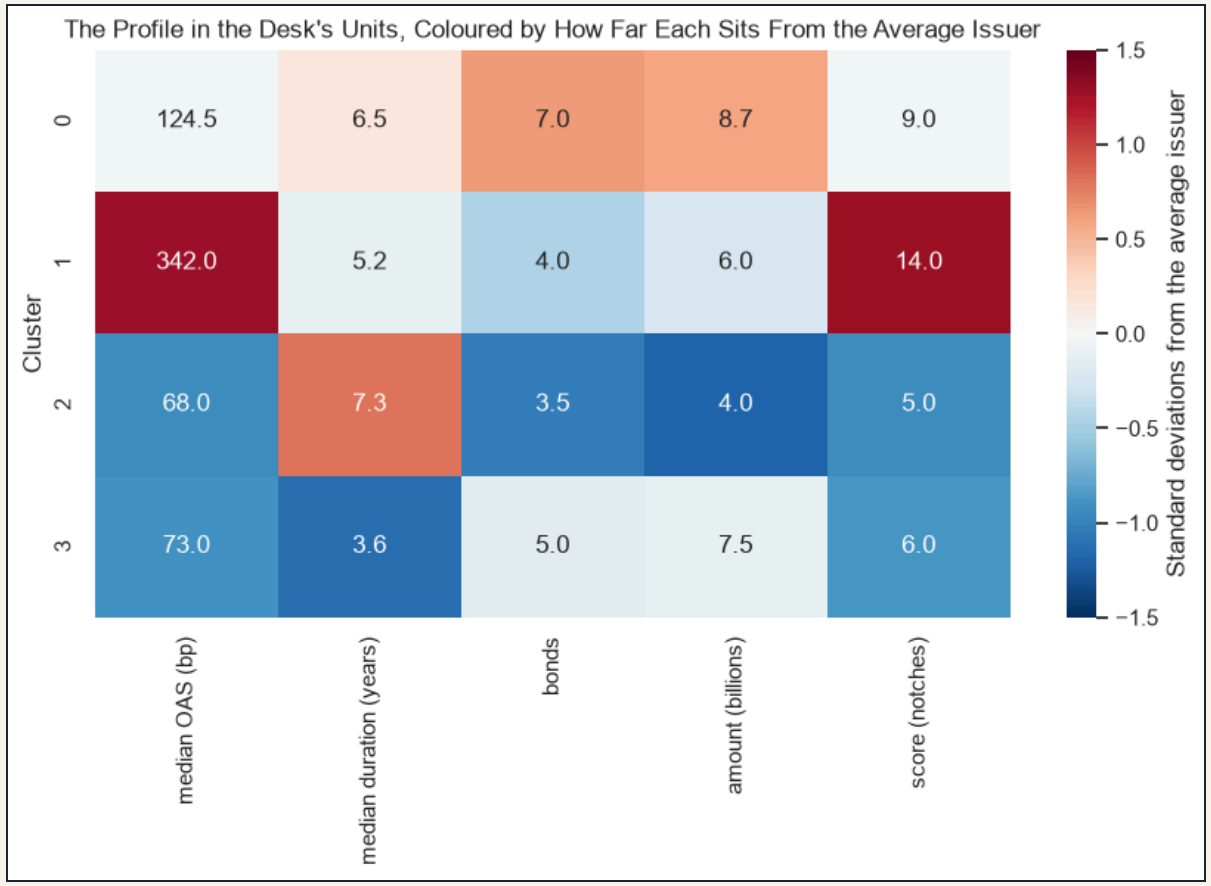
How Faithful Is the Tree



Each flat band is one merge: every pair it joins gets the same height.

Describe each cluster in bp, years and billions.

Profiling the Clusters



ONE LINE PER CLUSTER

```
def cluster_profile(rows, labels, columns):
    profile = rows.groupby(labels)[columns].median()
    profile["issuers"] = rows.groupby(labels).size()
    return profile
```

THE CLUSTERS IN WORDS

Cluster 1: 35 issuers, 342 bp, a median score of 14 (B+); 11.43 percent investment grade.

Each cluster now has a line a desk can read. Next: whether Day 12's K-means finds the same groups.

K-Means Against the Tree

K-MEANS ON THE SAME ISSUERS

```
kmeans = KMeans(n_clusters=4, n_init=10, random_state=SEED).fit(X_scaled)
issuers["kmeans"] = by_size(kmeans.labels_).to_numpy()
cluster_profile(issuers, issuers["kmeans"], units).round(2)
```

ITS PROFILE

KMEANS	MEDIAN_OAS	MEDIAN_DURATION	BONDS	AMOUNT_BN	ISSUERS
0	100.0	6.40	7.0	9.75	45
1	191.0	5.71	4.0	5.57	38
2	62.5	6.62	4.0	5.12	38
3	252.0	5.11	7.0	9.50	29

DROP THE RATING SCORE AND CLUSTER AGAIN

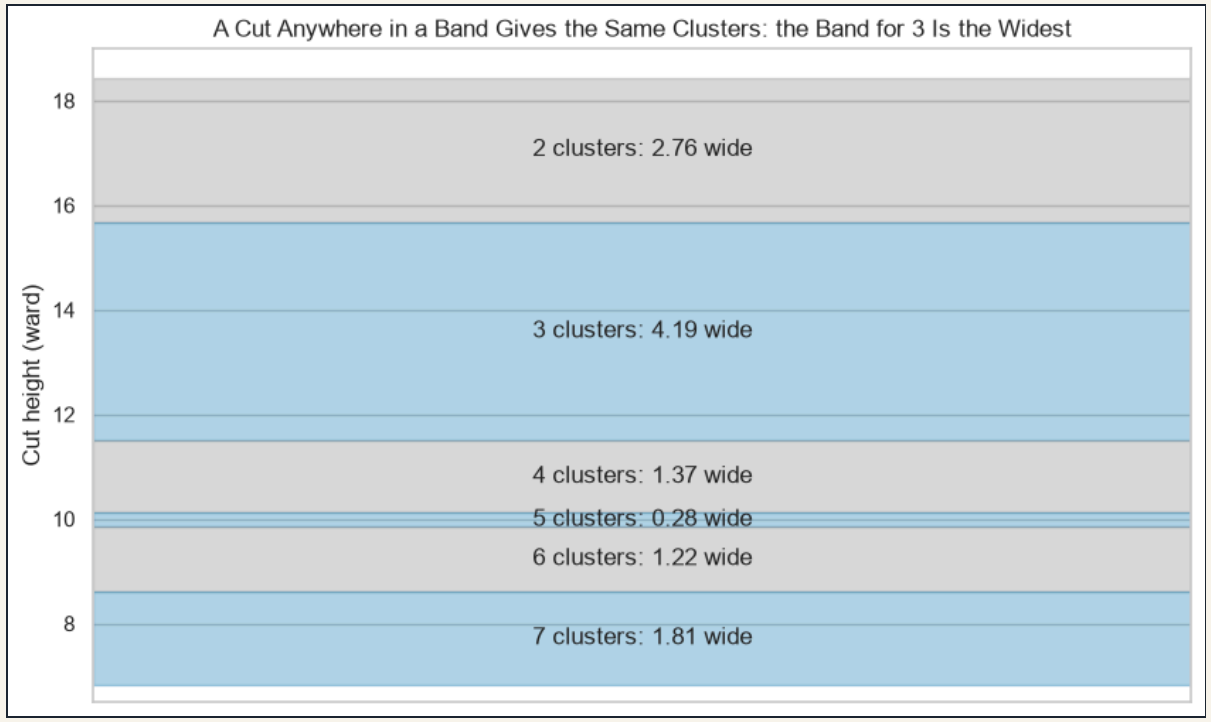
```
Sizes: [44, 42, 39, 25]
```

FOUR CLUSTERS BY K-MEANS

K-means makes four clusters of 45, 38, 38 and 29.

Two sound methods, two sets of groups: a clustering is one description of the issuers, not the answer.

Insights and Conclusion



HOW FAR THE CUT CAN MOVE

A cut anywhere from 11.49 to 15.68 gives 3 clusters: the widest band, 4.19.

The band for 4 is 1.37 wide, narrower than those for 2 and 7.

Day 12's silhouette barely moves, 0.197 to 0.231 for 2 to 10 clusters.

The tree comes first and the number of clusters after: one tree holds every cut.

You Can Now

- Build one row per issuer from its bonds with **`groupby`** and medians.
- Join rows from the bottom up with **`linkage`** and read the result as a dendrogram.
- Cut a tree with **`fcluster`**, by a number of clusters or a height, and say why you cut where you did.
- Compare linkages by their cluster sizes and their cophenetic correlation.
- Profile clusters in the file's own units, and set them beside K-means without ranking them.

NEXT
WEEK 3 · DAY 4

PCA and t-SNE

Day 14 squeezes many columns into a few with principal components.